Claire DAVID

Particle physicist • AI/ML educator • Science enabler

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French citizen, Canadian permanent resident

EDUCATION

2010 – 2016 | **PhD in Experimental Particle Physics, University of Victoria, Canada**
Supervisors: Rob McPherson & Michel Lefebvre
GPA: 8.63/9
Based at CERN (2012) for authorship qualification, ATLAS Experiment

2005 – 2010 | **Master of Engineering in Applied Physics, INSA Toulouse, France**
4th year: exchange semester on Fundamental Physics (in German), Humboldt-Universität Berlin
5th year: specialization in Nanophysics & Instrumentation
Rank: top 5%

APPOINTMENTS

2023 – 2025 | **Academic Director, *AI for Science**, AIMS South Africa**
Resident Researcher in Machine Learning, AIMS South Africa
* MSc program funded by Google DeepMind: ai.aims.ac.za

2022 – 2023 | **Visiting Scholar, AIMS South Africa**
McDonald Institute Ambassador
Canada's astroparticle physics network; sponsored fellow

2019 – 2023 | **Assistant Professor, York University, Canada**, jointly appointed as
Research Scientist, Fermilab, USA
ATLAS Experiment, CERN & Deep Underground Neutrino Experiment, Fermilab

2016 – 2019 | **Postdoctoral Researcher, DESY, Hamburg, Germany**
ATLAS Experiment, CERN

2010 – 2016 | **Graduate Research Associate, University of Victoria, Canada**
ATLAS Experiment, CERN; based at TRIUMF

PROFESSIONAL DEVELOPMENT

2026 | **Technical AI Safety, Blue Dot Impact**
6-week course with weekly facilitated discussions; certificate of completion

SKILLS

Research expertise: collider physics, machine learning, advanced data analysis, statistical inference

Programming: Python (NumPy, PyTorch, scikit-learn), C, C++, ROOT, LabVIEW, Unix shell

Communication: university-level instruction, public speaking, science popularization

Mentoring: academic coaching, student placement across international institutions

Languages: French (native), English (fluent), German (fluent), Spanish (highly proficient)

Contents: Research (p. 2) · Engineering Experience (p. 3) · Research Supervision (p. 3) · Teaching Experience (p. 4) · Student Placement (p. 6) · Academic Service (p. 7) · Grants & Awards (p. 9) · Presentations (p. 10) · Publications (p. 12) · Proceedings (p. 13) · Other Activities (p. 14)

RESEARCH

Highlights

2020 – 2022 | **Co-convener of Training & Documentation, DUNE Computing Consortium**
Deep Underground Neutrino Experiment, Fermilab; arXiv:2310.07342

2019 | **Co-founded DUNE-Canada group**
Deep Underground Neutrino Experiment, Fermilab

2018 | **Observation of the Higgs boson production mode with a top-antitop quark pair**
ATLAS Experiment, CERN LHC; Phys.Lett. B **784** (2018) 173

2016 – 2019 | **Led thermal validation and radiation length measurements of the ATLAS ITk silicon strip sensor support structure**
ATLAS Experiment, CERN LHC; ATLAS-TDR-025

2015 – 2016 | **pMSSM reinterpretation of LHC Run 1 SUSY searches; featured in CERN Courier**
ATLAS Experiment, CERN LHC; JHEP **10** (2015) 134

Current Projects

2025 – 2026 | **Automatic mapping of confidence sets to confidence intervals for intractable statistical models**
With Prof. Harrison B. Prosper (Florida State U.) & Prof. Olivia Prosper-Feldman (U. of Tennessee)
Expected completion: December 2026

As Resident Researcher in Machine Learning at AIMS South Africa

2025 – 2026 | **Physics-Informed Neural Network as solver of the photon orbit equation around a black hole**
With Tlotlo Oepeng (formerly AIMS South Africa) & Prof. H. B. Prosper (FSU)
Paper submitted to the Machine Learning for Physical Science (ML4PS) 2026 workshop

2025 | **Graph autoencoders for anomaly detection in particle physics**
With Raktim Mukherjee (Universität Heidelberg), Prof. H. B. Prosper (FSU)

As Research Scientist at Fermilab (York University, Toronto, Canada)

2020 – 2023 | **DUNE Computing Consortium**
Led first R&D on an interactive analysis facility for DUNE using columnar computing
Re-designed and co-organized three DUNE Computing tutorials → Front. Big Data **8** (2025) 1497622
Modernized DUNE Computing wiki; designed & deployed DUNE's automated glossary

2019 – 2023 | **DUNE-Canada group**
Secured 3-year NSERC funding to launch first Canadian effort (3 PIs)
Secured 5-year CFI infrastructure funding for a cryogenic facility for ArgonCube R&D

As Postdoctoral Researcher at DESY, Hamburg, Germany

2017 – 2019 | **Observation of $t\bar{t}h$ ($\rightarrow b\bar{b}$) process, ATLAS Experiment, CERN LHC**
Developed interface for Matrix Element Method in official analysis code
Validated Boosted Decision Trees' input distributions across Monte Carlo generators
Led fake lepton background estimate, one-lepton channel → Phys.Lett. B **784** (2018) 173

2016 – 2019

ATLAS Inner Tracker upgrade for High-Luminosity LHC
Led a team (3 engineers, 3 technicians, 4 students) to deliver first measurements of the prototype's thermal performance; developed infrared camera data acquisition interface
Led first measurements of radiation length (X/X0) using EUDET telescopes
Created custom entries of prototype's components in the production database

As Graduate Student Research Associate, University of Victoria, Canada

2013 – 2015

Search for electroweak Supersymmetry, ATLAS Experiment, CERN LHC
Main analyzer and editor of internal note of the search in the $h \to b\bar{b}$ channel
Reinterpreted the search as part of Run 1 summary paper in the context of the 19-parameter pMSSM
JHEP 10 (**2015**) 134

2012 – 2013

ATLAS Liquid Argon Calorimeter Operation
Software on-call expert during 2012 data taking (leading to the Higgs boson discovery)
Co-organized software expert training; created web-portal for shifters in control room
Developed ATLAS Online Software plugin revision to filter error messages

ENGINEERING EXPERIENCE

2010
5 months

ATLAS Forward Calorimeter upgrade, TRIUMF, Vancouver, Canada
Instrumented a radiation hardness test for diamond sensors using TRIUMF's cyclotron proton beam; created acquisition interface with TestPoint
Developed and tested the data analysis framework with ROOT → JINST **6** P05011 (2011)

2009
3 months

Pulse-shape modelling for IceCube, DESY, Berlin, Germany
Developed data analysis framework to measure the transfer function of IceCube Neutrino Observatory's photomultiplier modules, improving detector simulations

2008
2 months

Afterglow tests monitoring, Hilger Crystals Company*, UK
Monitored scintillation crystals afterglow tests under X-Rays; created a PhP/MySQL database for managing the test results
* developed crystals for the BaBar experiment

2007
1 month

French National High Magnetic Field Laboratory, Toulouse, France
Experimentations on the scotch-tape method for graphene exfoliation

RESEARCH SUPERVISION

At AIMS South Africa

2024 – 2026

Postdoc **Ryan Atkin**, AIMS South Africa
Main supervisor; anomaly detection for ATLAS High-Level Triggers

2025
2 months

MSc **Raktim Mukherjee**, visitor from Universität Heidelberg, Germany
Supervisor; investigation of Graph Auto-Encoder for anomaly detection for ATLAS triggers

At York University

2021 – 2023

Postdoc **Tejin Cai**, York University
Main supervisor; interactive analysis facility for DUNE on Compute Canada cloud
Now: software engineer at Black Duck

2020

MSc **Alfred Ferwerda**, York University
Thesis committee member; CERN ALPHA Experiment

2019 – 2020 | PhD **Darij Starko**, York University
Local supervisor; ATLAS qualification task, Flavour tagging group, CERN
Now: data science consultant at Wyatt AI Inc.

2021 | BSc **Amin Jafarov**, York University, NSERC USRA recipient
Main supervisor; searches for long-lived particles using ML
Now: Medical School, University of Central Lancashire, UK

2021 | BSc **Anita Nasiri**, York University
Co-op research project supervisor; long-lived particle event 3D visualisation in ATLAS

2020 | BSc **Majd Salam**, York University
Co-op research project supervisor; first ML classifier for ATLAS long-lived particle search
Now: PhD candidate, Theoretical Astrophysics, Western University, Canada

2020 | BSc **Damith Deshan**, York University
Co-op research project supervisor; automatic scans of chip pack labels for ATLAS ITk production database

As postdoctoral researcher at DESY

2016 – 2019 | PhD **Jan-Hendrik Arling**, DESY
Local technical supervisor; ATLAS ITk upgrade
Now: junior staff at DESY

2018 | BSc **Anna Beck**, DESY summer student
Supervisor; ATLAS ITk thermal characterization
PhD at University of Warwick; now postdoctoral researcher at MIT

2017 | BSc **Maximilian Caspar**, DESY summer student
Supervisor; ATLAS ITk upgrade, instrumentation
PhD at Universität Wuppertal/DESY; now consultant at Basycon

2016 – 2017 | PhD **Yasiel Delabat Diaz**, DESY
Local supervisor; ATLAS qualification task, ITk upgrade
PhD at Humboldt-Universität/DESY; now software developer at Safetec

As research assistant at University of Victoria

2011 | BSc **Philipp Koenig**, University of Victoria
Co-op project co-supervisor; ATLAS Higgs group
PhD at Universität Bonn; now senior data analyst at Syna

TEACHING EXPERIENCE

Invited Lectures

Main course website: clairedavid.github.io/intro_to_ml

Oct. 2025 | **Introduction to Artificial Intelligence & Data Science, AIMS Cameroon**
Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 108 students
Skill course, required for all MSc streams

Sep. 2025 | **Introduction to Machine Learning (modernized version), AIMS South Africa**
MSc course *AI for Science*; 10 lectures (20 hours) + tutorials (5 hours); 33 students
Co-teaching with Prof. Harrison Prosper (FSU)

Jul. 2025	**Basic Concepts in Machine Learning, EASY-ML* Summer School, Nairobi, Kenya** 10 lectures (15 hours) + tutorials (5 hours); 35 students * East-African School for Young Researchers on Advanced Machine Learning; CNRS, France
Sep. 2024	**Introduction to Machine Learning, AIMS South Africa** MSc course *AI for Science*; 15 lectures (30 hours) + tutorials (10 hours); 21 students Delivered again in December–January for 10 students who joined late due to visa delays
Apr. 2024	**Machine Learning by Hand, AIMS South Africa** MSc course *AI for Science*; 15 lectures (30 hours) + tutorials (10 hours); 14 students Adapted content for students joining in January due to visa delays
Feb. 2024	**Introduction to Artificial Intelligence & Data Science, AIMS Cameroon** Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 35 students
Jan. 2024	**Introduction to Machine Learning, AIMS Senegal** Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 20 students
Sep. 2023	**Introduction to Machine Learning, AIMS South Africa** MSc course *AI for Science*; 15 lectures (30 hours) + tutorials (10 hours); 16 students + 24 auditing students from the Mathematical Science stream
May 2023	**Introduction to Machine Learning & Data Science, AIMS Ghana** Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 19 students
Mar. 2023	**Learn the Math behind Neural Networks, University of Johannesburg, South Africa** One-day workshop: theory (3 hours) and practice in Python (3 hours); 22 participants Open to all academic levels from 4th year undergraduate
Feb. 2023	**Machine Learning in Particle Physics, AIMS South Africa** Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 33 students
Jan. 2023	**Machine Learning in Particle Physics, AIMS Senegal** Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 19 students
Dec. 2022	**Introduction to Machine Learning, ASP2022*, Gqeberha, South Africa** Two morning sessions (6 hours); 40 participants * African School of Physics: africanschoolofphysics.org
Jan. 2019	**Life of a Particle: Stochastic Modelling in Particle Physics, AIMS Ghana** Graduate course; 15 lectures (30 hours) + tutorials (10 hours); 38 students Co-taught with Dr. Samuel Meehan

Teaching at York University

Fall 2021	**Special Relativity and Introduction to Modern Physics (PHYS2040)** Second year undergraduates in Physics & Engineering Majors (online); 57 students
Fall 2020	**Special Relativity and Introduction to Modern Physics (PHYS2040)** Second year undergraduates in Physics & Engineering Majors (online); 64 students
Fall 2019	**Special Relativity and Introduction to Modern Physics (PHYS2040)** Second year undergraduates in Physics & Engineering Majors; 58 students

Previous Teaching Experience

Sep. 2018	**Autumn School for High Energy Physics, Maria Laach, Germany** Invited group coordinator; tutoring graduate students projects in experimental particle physics; 15 participants

Jan. 2016 — **TRIUMF Junior Research School, Vancouver, Canada**
Invited lecturer: "The Higgs boson mechanism"; 10 Singaporean college students

Jan. 2015 — **TRIUMF Junior Research School, Vancouver, Canada**
Invited lecturer: "The Higgs boson mechanism"; 10 Singaporean college students

STUDENT PLACEMENT

As Academic Director at AIMS South Africa

Matched students with capstone supervisors; one-on-one mentoring & tailored career guidance; connected and referred students for post-AIMS positions.

2024 – 2025 — **33 students guided to MSc graduation, *AI for Science***
Selected placements:
Ahmed Nadir, Research Intern, ML & Global Health Network, University of Oxford; active learning for survey design
Antony Gitau, PhD student, University of South-Eastern Norway; AI for medical imaging
Benedict Ejelonu, G-RIPS Fellow, Tohoku University, Japan; AIMS–Tübingen Data Science & AI Fellow
Harlette Denebeye, Marie Skłodowska-Curie Doctoral Researcher, Université Paris-Saclay/CEA-CNRGH; trustworthy AI
Josiah Kimani, AI for Weather Prediction researcher, AfriClimate AI; 2nd prize poster, Deep Learning Indaba 2025
Koomi Toussaint Amoussouvi, AI for Weather Prediction researcher, AfriClimate AI; 1st prize poster, Deep Learning Indaba 2025
Mushtaha El-Amin, Visiting Researcher, LiGHT Laboratory, EPFL; global health & humanitarian response technologies
Omer Ebead, Research Fellow, Cooperative AI Foundation; multi-agent privacy; intern at InstaDeep on MARL
Onintsoa Rakotondranisa, PhD student, University of Cape Town; astrophysics
Rodeo Toha, Head of data and AI at Grow-UP AI, Abomey-Calavi, Benin
Tim Hitge, Research Assistant, Imperial College London; surrogate models & simulation-based inference; paper + poster at NeurIPS 2025

2023 – 2024 — **30 students guided to MSc graduation, *AI for Science* inaugural cohort**
Selected placements:
Abel Legese, Senior AI Research Engineer, Ersilia; open-source drug discovery
Carlin Foka, PhD student, Centre for Epidemic Response & Innovation (CERI)
Catherine Essuman, PhD candidate (from Sept. 2026), Ontario Tech University, Canada; Canada Impact+ Research Award recipient
Daniel Rajaonarivelomanantsoa, Research Fellow, InstaDeep; first Malagasy oral presenter at NeurIPS 2025; PhD, Stellenbosch University
Dariose Diffo, PhD student, Università di Bologna; AI & computational methods for astronomy and physics
Haingo Andry, Recipient of the 2025 Google PhD Fellowship, Stellenbosch University/CERI
Herman Tesso, Doctoral Researcher, Aalto University, Finland; first-author paper at AABI workshop, ICLR 2025; simulation-based inference & spatial statistics
Isaac Omolayo, PhD student, Universität Tübingen; contrastive learning for connectome-constrained neural networks, fruit fly visual system
Shâma Wache, PhD student, TU Chemnitz, Germany; inverse problems in medical imaging
Tiana Ramonjison, MSc student, Simon Fraser University/TRIUMF; dark matter detector design

Informal Mentoring & Connections

2025 — **Catherine Essuman & Pelumi Aderinto**, *AI for Science* alumni
Led on my behalf the tutorial "The Math Behind Neural Networks" at Deep Learning Indaba, Lagos, Nigeria

2025 — **Angel Bashyal**, MSc in Astrophysics, Tribhuvan University, Nepal
Met at BCVSPIN 2024; connected to potential supervisors via McDonald Institute network
Recipient of Swiss Government Excellence Scholarship; PhD in Astrophysics, start Sept. 2026

2025 — **Nishan Lamichhane**, MSc student, Tribhuvan Vishwavidalaya, Nepal
Connected with Prof. Aaron Vincent, Queen's University, for honour's research project
Now accepted for a PhD in Physics & Astronomy at Queen's University; start Sept. 2026

2024 — **Jeffrey Katabo**, MSc student, AIMS Cameroon
Connected with Prof. Joel Zylberberg (CIFAR Fellow, York University); meta-learning for models of primate visual cortex (AIMS capstone project)
Now pre-doctoral research fellow, Freie Universität Berlin / Bernstein Center for Computational Neuroscience

2023 — **Ufuk Çakır**, AIMS–Universität Heidelberg research intern
DPhil student, Intelligent Earth CDT, University of Oxford; RL for wildfire response

ACADEMIC SERVICE

Service to AIMS

2024 – 2026 — **Designed and led student onboarding programme, AIMS South Africa**
Week-long induction: campus life, values, systems, cultural exchange, wellness walks
4 editions: Sept. 2024, Jan. 2025, Sept. 2025, Jan. 2026

2023 – 2025 — **Founded student clubs to foster initiative and campus life, AIMS South Africa**
Student-run, with dedicated budget; adapted each year to cohort interests
2023–24: Arts, Fitness, Hike & Walk, Paper Reading, Running
2024–25: Board Game, Cultural, Dance, Media, Running, Soccer

2023 – 2025 — **Co-created formal lecturer contracts and tutor agreements, AIMS South Africa**

Service to Fermilab/York University

2020 – 2021 — **Member, Fermilab Neutrino Division postdoc hiring committee**

2020 – 2021 — **Co-chair, Physics & Astronomy Department Weekly Colloquium, York University**

2019 – 2020 — **Member, Faculty of Science Award Committee, York University**

Service to the Field

2024 — **Session co-chair, Sustainability Track, 42nd International Conference on High Energy Physics (ICHEP), Prague**
First dedicated sustainability track at ICHEP; environmental impact of HEP accelerators, detectors and computing

2022 – 2023 — **Sustainable HECAP+ Initiative**
Creator of sustainable-hecap-plus.github.io; advocate for sustainable practices in high-energy physics

2022 — **Judge, Best Student Competition, Particle Physics division**
Canadian Association of Physicists' Congress, Hamilton, Canada

2021 – 2022	**Elected Vice-Chair, Particle Physics Division, Canadian Association of Physicists (CAP)**
2021	**Session convener, Neutrino session, CAP Congress 2021**
2020 – 2022	**Co-convener, DUNE Training & Documentation working group, Fermilab**
2019 – 2020	**Institute contact (York University), ATLAS Inner Tracker production database**

Memberships

2024 – 2025	Hydrogen Intensity and Real-time Analysis eXperiment (HIRAX) Member of Management Team
2023 – 2026	South African Institute of Physics (SAIP)
2020 – 2022	Canadian Association of Physicists (CAP)
2019 – 2022	Institute of Particle Physics (IPP), Canada

Peer Review

2026	**Reviewer, Nederlandse Organisatie voor Wetenschappelijk Onderzoek (NOW)** Dutch Research Council
2025	**Nominated reviewer, Machine Learning for Physical Sciences Workshop (ML4PS), NeurIPS 2025** Reviewed 2 papers
2024	**Nominated reviewer, Machine Learning for Physical Sciences Workshop (ML4PS), NeurIPS 2024** Reviewed 2 papers
2019	**Referee, Journal of Instrumentation** Nominated by peers
2017	**Referee, Journal of Instrumentation** Nominated by peers

Editorial Work

2021	**Selected reviewer, DUNE Vertical Drift Conceptual Design Report** Chapter: Trigger & Data Acquisition
2019 – 2021	**Editorial Board Member, ATLAS Collaboration** "Configuration and performance of the ATLAS b-jet triggers in Run 2"; Eur. Phys. J. C **81** (2021) 1087

Event Organization

2025	**International Masterclass "Hands-on Particle Physics", AIMS South Africa** First edition hosted by AIMS; 25 high school students from 8 Cape Town schools Organized in partnership with IPPOG; live conference with CERN
2023 – 2024	**Siyakhula Festival, AIMS 20th Birthday Celebration, Muizenberg, South Africa** Organizing committee member; 6-day festival, March 17–22, 2024
2021	**DUNE Computing Tutorial, extended version (online)** Co-organizer; 3-day format with interactive platform, quizzes and expert Q&A

2021 | **Poster session organizer, TIPP 2021 (online)**
International Conference on Technology and Instrumentation in Particle Physics; GatherTown platform

2021 | **DUNE Computing Tutorial, one-day workshop (online)**
Main organizer; teaching basics of DUNE computing to new members

2019 | **ATLAS Higgs & Top Quark Workshop (HTop), DESY, Hamburg, Germany**
Member of the Local Organizing Committee

Other Service

2007 – 2008 | **Founder & spokesperson, Commission Réussite, INSA Toulouse**
Designed & implemented programme where students in difficulty choose a mentor among professors
Approved by administration as a dedicated commission; system still active in 2026

GRANTS & AWARDS

Research Grants

2022 – 2027 | **Canadian Foundation for Innovation (CFI) John R. Evans Leaders Fund (JELF)** (PI, CAD $125,000)
Next-generation neutrino detector for DUNE

2022 – 2023 | **Compute Canada Fast Track Competition, one-year extension** (PI, CAD $23,339)

2021 – 2022 | **Compute Canada Resources Competition** (PI, CAD $32,225)
Interactive Analysis Facility for DUNE

2021 – 2022 | **NSERC Discovery COVID-19 Extension** (non-competitive, CAD $50,000)

2020 – 2023 | **NSERC Subatomic Physics Discovery Grant** (Co-PI, CAD $465,000)
Neutrinos exposed: from the cosmos to deep underground

2020 – 2021 | **NSERC Subatomic Physics Discovery Grant** (PI, CAD $50,000)
Higgs boson and top quark in ATLAS: gateway to new physics

2019 – 2023 | **York University Internal Research Fund, Startup** (CAD $110,000)

Awards & Recognitions

2025 | **Breakthrough Prize in Fundamental Physics**
Shared with ATLAS, CMS, ALICE and LHCb collaborations (13,508 researchers)
breakthroughprize.org

2020 | **York University Minor Research Grant & Junior Faculty Fund** (CAD $3,100)

2017 – 2018 | **DESY Bonus Merit** (€3,200)
Thermal validation of ATLAS ITk prototype

2016 – 2017 | **DESY Bonus Merit** (€3,200)
First infrared measurements of ATLAS ITk sensor structures

2016 | **DESY Postdoctoral Fellowship**, High Energy Physics division

2011 – 2012 | **Nora and Mark de Goutiere Memorial Fellowship** (CAD $10,000)

2010 | **University of Victoria Entrance Award** (CAD $10,000)
Nominated for Vanier Scholarship, Dept. of Physics & Astronomy, UVic

PRESENTATIONS

International Conferences

2026 **High Dimensional Problems for Statistical Methods in Fundamental Physics, BIRS, Banff, Canada**
Invited talk: "Leveraging Machine Learning, and its limits" (co-presenter: Anthony Ozerov)

2025 **32nd International Workshop on Deep Inelastic Scattering (DIS2025), Cape Town, South Africa**
Invited plenary talk: "Diversity, Equity and Inclusion in Physics"

2023 **11th Large Hadron Collider Physics Conference (LHCP), Belgrade, Serbia**
Invited talk: "Statistics on participation of emerging countries in LHC and other experiments"

2020 **30th International Linear Accelerator Conference (LINAC), Liverpool, UK (online)**
Invited panelist: "Women in Science and Engineering: Should I stay or should I go now?"

2019 **Conference on Flavor Physics and CP Violation (FPCP), Victoria, Canada**
Invited talk: "Higgs boson couplings to quarks at the ATLAS experiment"

2018 **Conference on Standard Model at LHC, Berlin, Germany**
Young Scientist talk: "Search for $t\bar{t}H$ production in the $H \to b\bar{b}$ channel"

2017 **11th International Conference on Position Sensitive Detectors (PSD11), Milton Keynes, UK**
Invited plenary talk: "A new strips tracker for the upgraded ATLAS ITk detector"

2016 **Lake Louise Winter Institute, Alberta, Canada**
Invited plenary talk: "Search for supersymmetry with the ATLAS detector in final states with leptons, jets and missing transverse momentum"

2014 **International Conference on High-Energy Physics (ICHEP), Valencia, Spain**
Poster: "Search for chargino and neutralino production with a Higgs boson in the decay chain with one or three leptons final state events"

2013 **International Conference on New Frontiers in Physics (ICNFP), Kolymbari, Greece**
Invited talk: "ATLAS results in SUSY searches for 3rd generation squarks and electroweak production"

Funding Agencies & Institutional Reviews

2021 **Institute of Particle Physics (IPP) Annual General Meeting (online)**
Invited talk: "Deep Underground Neutrino Experiment: current status, Canadian effort & funding"

2018 **DESY Program-Oriented Funding (POF) Evaluation, Hamburg, Germany**
Invited poster: "ATLAS and CMS tracker end-caps for the High-Luminosity LHC" (co-author Dr. J. Keaveney); official tour guide for Helmholtz Association reviewers

2016 **Helmholtz Programme Matter and the Universe, Mainz, Germany**
Invited poster: "Anatomy of ATLAS and CMS strip tracker end-caps for High-Luminosity LHC"

2013 **TRIUMF International Review Committee, Canada**
Invited poster: "Search for Higgs production in supersymmetric decay cascades with ATLAS"

2013 **VIP Visit, ATLAS Control Room, CERN, Geneva, Switzerland**
Invited speaker for P. Young (Chair, TRIUMF Board of Management) & G. Patry (CEO, Canada Foundation for Innovation)

Local & Regional Conferences

2021 **Canadian Association of Physicists Congress (CAP), online**
Invited talk: "Deep Underground Neutrino Experiment: Review and Recent Progress"

2021 **Third Summer Particle Astrophysics Workshop, Queen's University, Canada (online)**
Plenary talk: "The Deep Underground Neutrino Experiment: DUNE"

2020 **Winter Nuclear & Particle Physics Conference (WNPPC), Banff, Canada**
Plenary talk: "Higgs boson physics and upgrade work with ATLAS at the Large Hadron Collider"

2019 **HTop Workshop, DESY, Hamburg, Germany**
Plenary talk: "Small backgrounds in the leptonic channels"

2018 **Preliminary Design Review ITk Strip Local Support Structures, CERN**
Final presentation of results on thermo-mechanical petal prototypes; passing review

2018 **Forum on Tracking Detector Mechanics, Valencia, Spain**
Plenary talk: "Thermo-mechanical characterization of Petals as Local Support Structures for the ATLAS ITk Strip Detector"

2018 **HTop Workshop, CERN, Switzerland**
Invited talk: "Fakes and other backgrounds in $t\bar{t}H(b\bar{b})$ leptonic channels" (co-author J. Mellenthin)

2014 **LHCC, CERN, Switzerland**
Poster: "Search for charginos, neutralinos and sleptons in one or two leptons final state events with ATLAS" (co-author R. Ueno)

2011 **ATLAS Canada Workshop, Vancouver, Canada**
Talk: "Search for SUSY Higgs in the $b\bar{b}$ final state"

Public Talks

2026 **Career Thursday, Qhubeka Youth Development Programme, Langa, South Africa**
Featured guest; Q&A on experimental physics & AI in education; organized by Just Grace

2023 **Université Kongo, Mbanza-Ngungu, Congo DRC**
Co-talk with Dr. Nour Skaf: "From the smallest to the largest: a journey from particle physics to astrophysics"

2023 **Dakar Academy, Senegal**
Invited speaker: experimental particle physics to 200+ college-preparatory students

2022 **Neutrino Day, Sanford Underground Research Facility, Lead, South Dakota, USA**
Keynote speaker at Lead's Opera House; VIP underground visit with YouTuber Physics Girl (Dianna Cowern)
Live radio broadcast for South Dakota radio about DUNE and Neutrino Day

2021 **Royal Astronomical Society of Canada, Toronto, Canada**
Invited speaker, Speaker's Night livestream: "What is DUNE? Embark on the underground telescope to pierce the secrets of matter!" [YouTube]

2015 **Science Innovation & Technology Speaker Series, Richmond, BC, Canada**
"The Triumphant Discovery of the Higgs Boson"

2014 **Cafés Scientifiques, Alliance Française, Vancouver, Canada**
"La triomphante découverte du boson de Higgs"; organized by the French Consulate

Invited Colloquia

2026 **Integrative Think Tank (ITT), AIMS South Africa**
Five-day industry-academia workshop; invited talk: "How I became a scientist: the backstory not read on my CV"

2025 **AIMS Cameroon, Limbé, Cameroon**
"A journey inside matter: experimental particle physics with ATLAS and DUNE"

2025 **AIMS Cameroon, Limbé, Cameroon**
"How I became a scientist: the backstory not read on my CV"

2024 **AIMS Cameroon, Limbé, Cameroon**
"A journey inside matter: experimental particle physics with ATLAS and DUNE"

2024 **AIMS Cameroon, Limbé, Cameroon**
"How I became a scientist: the backstory not read on my CV"

2023 **Université de Montréal, Département de Physique, Canada**
"Half a year in Africa: AI, AIMS & opportunities"

2023 **East African Institute for Fundamental Research (EAIFR), Kigali, Rwanda**
Co-talk with Dr. Nour Skaf: "From the smallest to the largest: a journey from particle physics to astrophysics"

2023 **University of Ghana, Department of Physics, Accra, Ghana**
"A journey inside matter: experimental particle physics with ATLAS and DUNE"

2023 **AIMS Rwanda, Kigali, Rwanda**
"A journey inside matter: experimental particle physics with ATLAS and DUNE"

2023 **AIMS Cameroon, Limbé, Cameroon**
"A journey inside matter: experimental particle physics with ATLAS and DUNE"

2023 **Université de Lomé & CERME, Lomé, Togo**
"The ATLAS Experiment at CERN: a detector, an analysis and an upgrade program"

2023 **International Chair in Mathematical Physics & Applications, Cotonou, Benin**
"A journey inside matter: experimental particle physics with ATLAS and DUNE"

2023 **AIMS South Africa, Muizenberg, South Africa**
"How I became a scientist: the backstory not read on my CV"

2020 **McGill Physical Society Colloquium, Montréal, Canada (online)**
"Data analysis and detector development with ATLAS at the Large Hadron Collider" [YouTube]

2019 **York University, Dept. of Physics & Astronomy Colloquium, Toronto, Canada**
"Sorting needles in a haystack and cooling a sandwich: data analysis and detector development with ATLAS"

PUBLICATIONS

2026 **David C.**, Oepeng T., Prosper H., "PINN-Slicer: Exploiting Rotational Symmetry to Solve the Photon Orbit Equation Around a Black Hole"
Submitted to the ML4PS 2026 workshop

2025 Reinsvold Hall A., Skidmore N., Benelli G., Carlson B., **David C.** et al., "Training and onboarding initiatives in high energy physics experiments"
Front. Big Data **8** (2025) 1497622

2025 Sustainable HECAP+ Initiative, "Environmental sustainability in basic research: a perspective from HECAP+"
JINST **20** (2025) P03012; www.hecap.eco

2022 DUNE Collaboration, "Design, construction and operation of the ProtoDUNE-SP Liquid Argon TPC"
JINST **17** (2022) P01005

2021 ATLAS Collaboration, "Search for exotic decays of the Higgs boson into long-lived particles in pp collisions at $\sqrt{s} = 13\,\text{TeV}$ using displaced vertices in the ATLAS inner detector"
JHEP **11** (2021) 229

2021 DUNE Collaboration, "Deep Underground Neutrino Experiment (DUNE) Near Detector Conceptual Design Report"
Instruments **5** (2021) 31

2020 Poley L., Sawyer C. et al., "The ABC130 barrel module prototyping programme for the ATLAS strip tracker"
JINST **15** (2020) P09004

2020 DUNE Collaboration, "Long-baseline neutrino oscillation physics potential of the DUNE experiment"
Eur. Phys. J. C **80** (2020) 978

2018 ATLAS Collaboration, "Observation of Higgs boson production in association with a top quark pair at the LHC with the ATLAS detector"
Phys. Lett. B **784** (2018) 173–191

2018 ATLAS Collaboration, "Search for the SM Higgs boson produced in association with top quarks and decaying into a $b\bar{b}$ pair in pp collisions at $\sqrt{s} = 13\,\text{TeV}$ with the ATLAS detector"
Phys. Rev. D **97** (2018) 072016

2018 Albertsson K. et al., "Machine Learning in High Energy Physics Community White Paper"
J. Phys.: Conf. Ser. **1085** (2018) 022008

2017 ATLAS Collaboration, "Technical Design Report for the ATLAS Inner Tracker Strip Detector"
CERN-LHCC-2017-005 ; ATLAS-TDR-025

2015 ATLAS Collaboration, "Summary of the ATLAS experiment's sensitivity to supersymmetry after LHC Run 1 — interpreted in the phenomenological MSSM"
JHEP **10** (2015) 134

2015 ATLAS Collaboration, "Search for direct pair production of a chargino and a neutralino decaying to the 125 GeV Higgs boson in $\sqrt{s} = 8\,\text{TeV}$ pp collisions with the ATLAS detector"
Eur. Phys. J. C **75** (2015) 208

2013 ATLAS Collaboration, "Search for chargino and neutralino production in final states with one lepton, two b-jets consistent with a Higgs boson, and missing transverse momentum with the ATLAS detector in $20.3\,\text{fb}^{-1}$ of $\sqrt{s} = 8\,\text{TeV}$ pp collisions"
ATLAS-CONF-2013-093

2011 Axen D. et al., "Diamond detector irradiation tests at TRIUMF"
JINST **6** (2011) P05011

PROCEEDINGS

2025 **David C.**, "What is AI, what is it not, how we use it in physics and how it impacts... you"
BCVSPIN 2024: Particle Physics and Cosmology in the Himalayas, Kathmandu, Nepal
arXiv:2504.01827

2024 **David C.**, "Statistics on participation of emerging and developing countries in LHC and other experiments"
PoS(LHCP2023) **450** (2024) 062

2018 **David C.**, "A new strips tracker for the upgraded ATLAS ITk detector"
Proceedings of PSD11
JINST **13** (2018) C01003

2016 **David C.**, "Search for chargino and neutralino production with a Higgs boson in the decay chain in one or three leptons final state events with ATLAS"
Nucl. Part. Phys. Proc. **273–275** (2016) 2400–2402

2014 **David C.**, "New ATLAS results in SUSY searches for 3rd generation squarks and electroweak production"
EPJ Web of Conferences **71** (2014) 00034

OTHER ACTIVITIES

Panels

2025 **SA-CERN 15 Year Celebrations, iThemba LABS, South Africa**
Invited panelist: "Fundamental Science for South African Development"

2023 **Science & Technology Week, Kinshasa, Congo DRC**
Invited panelist: "Science and technology for sustainable development in Africa" (in French)

2023 **11th Large Hadron Collider Physics Conference (LHCP), Belgrade, Serbia**
Invited panelist: "Challenges for emerging countries to join/participate in LHC experiments"

2020 **30th International Linear Accelerator Conference (LINAC), Liverpool, UK (online)**
Invited panelist: "Women in Science and Engineering: Should I stay or should I go now?"

2014 **Screening of "Particle Fever", Vancouver International Film Festival, Canada**
Invited panelist with movie producer & Prof. David Kaplan

2011 **Emily Carr University of Art and Design, Vancouver, Canada**
Panel speaker: "But $\hbar$? Particle Interactions in Physics and Arts: Artistic Transformation"

Science Festivals & Tours

2023 **Village of Sciences, École de la Gombé, Kinshasa, Congo DRC**
Introduced CERN and ATLAS to 10,000+ high school students; trained selected learners as booth animators

2017 **DESY Open Day (DESYDAY), Hamburg, Germany**
Organizer of ATLAS exhibition & infrared camera booth; 20,000+ visitors

2015 – 2016 **TRIUMF Tour Guide, Vancouver, Canada**
Monthly tours for high schools, graduate students and the general public

2014 **ATLAS Masterclass, University of Victoria, Canada**
One-day hands-on for high-school students; invited talk: "Data analysis with ATLAS at the LHC"; organized by VISPA

2013 **TRIUMF Open House, Vancouver, Canada**
Volunteer at the ATLAS booth; 1,200+ visitors

Online & Media

2023 **ATLAS Virtual Visit, ATLAS Control Room, CERN**
Lycée Liziba, Kinshasa, Congo DRC; girls' secondary school [atlas.cern]

2023 **ATLAS Virtual Visit, ATLAS Control Room, CERN**
Malaika School, Lubumbashi, Congo DRC; free education for 400+ girls [atlas.cern]

2021 **Allan I. Carswell Observatory, York University, Canada**
TeleTube series: "The Triumphant Discovery of the Higgs Boson at the Large Hadron Collider" [YouTube]

2020 **"The Drift" series, Sanford Underground Research Facility, USA**
"What is so special about neutrinos?" on SURF's YouTube channel [YouTube]

2020 **Neutrino Day, SURF's Free Science Festival (online)**
"DUNE scientists answer your questions" [YouTube]

2013 **ATLAS Virtual Visit, ATLAS Control Room, CERN**
Europe Day Symposium, Vancouver; TELUS World of Science Celebration, Vancouver [video]